

Project Logbook

Phase 1 — Project Ideation & Planning (Late January – Early February)

- Project initiated in late January 2026
- Initial goal was to create a manufacturable embedded environmental monitoring system capable of measuring temperature, humidity, and atmospheric pressure
- Early concepts explored both Raspberry Pi and standalone embedded approaches before committing to a custom PCB architecture
- Selected the BME280 as the primary environmental sensor due to its ability to measure temperature, humidity, and atmospheric pressure in a single package
- Established GitHub repository for version control and organization
- Created the initial project definition document outlining objectives and proposed functionality

Phase 2 — Proof of Concept & Sensor Validation (February – Early March)

- Used a Raspberry Pi temporarily as a proof-of-concept platform to validate environmental sensing and data collection
- Initial BME280 units were damaged during testing due to incorrect pin labeling and wiring confusion
- Multiple communication and wiring configurations were tested while troubleshooting sensor reliability
- DHT22 was temporarily used to validate the sensing workflow while replacement BME280 units were obtained
- Basic GUI and sensor visualization concepts were tested during this stage
- Proof-of-concept testing confirmed the feasibility of the environmental monitoring system before transitioning to standalone PCB development

Phase 3 — Transition to Custom PCB Architecture (March 5th)

- During design review it became clear that a Raspberry Pi and touchscreen-based system would not reflect a realistic manufacturable embedded product
- Team agreed that a standalone embedded PCB design would better align with course goals and real engineering practices
- Decision made to redesign the system around the ATmega328P-A microcontroller

- Replaced the touchscreen GUI concept with USB serial communication and future phone notification capability
- Replacement BME280 sensor was correctly wired and validated, officially replacing the DHT22 proof-of-concept setup
- Project scope shifted toward a manufacturable PCB-focused embedded system

Phase 4 — Schematic Design & Embedded Hardware Development (March – April)

- Schematic designed in KiCad using the ATmega328P-A as the primary processing unit
- Integrated the BME280 environmental sensor for temperature, humidity, and atmospheric pressure monitoring
- Used the available KiCad BME280 footprint and configured the sensor for I2C communication
- External crystal oscillator and 22pF load capacitors added to improve MCU timing stability
- Selected passive component values directly from component datasheets
- Added decoupling capacitors to reduce electrical noise
- Added ISP programming header for direct firmware flashing
- Added test points for probing, debugging, and future expansion
- Added status LED with current-limiting resistor for visual debugging

Phase 5 — PCB Layout & Design Review (April 1st)

- Initial PCB layout completed and reviewed collaboratively
- Design review identified missing decoupling and bulk capacitors which were later added
- Routing constraints on a standard 2-layer PCB created congestion and limited trace flexibility
- Experimented with converting the design to a 4-layer stackup to learn inner-layer routing and via assignment techniques
- Learned how to assign vias to specific layers and route signals across internal copper layers
- ERC and DRC checks were performed and violations were resolved
- Generated 3D board renders to validate physical layout and component placement

Phase 6 — Manufacturing Preparation (Late April – Early May)

- Generated Gerber fabrication files for all PCB layers
- Exported drill files for PCB manufacturing
- Compiled BOM containing component values, quantities, and part numbers
- Prepared PCB ordering instructions for fabrication services such as JLCPCB and PCBWay

- Organized repository structure with fabrication outputs, documentation, and software resources
- Verified the project was manufacturing-ready and capable of future assembly

Phase 7 — Final System Integration & Presentation Preparation (May 7th)

- Updated the design to use two dedicated USB-C connectors — one for board power and one for communication with external software
- Began transitioning away from older screw terminal-only power concepts toward a cleaner and more modern interface design
- USB serial communication workflow finalized between the ATmega328P-A and external computer application
- Python-based data visualization and future weather trend prediction concepts integrated into the final presentation workflow
- Discussed future implementation ideas involving ntfy notifications, wireless monitoring, and predictive weather analysis
- Final presentation prepared demonstrating the complete engineering workflow from concept to manufacturable PCB
- Project finalized and submitted on May 7th, 2026

Lessons Learned

- Hardware reliability must always be verified before full system integration
- Incorrect sensor pin labeling can quickly damage components and delay development
- Early proof-of-concept platforms are useful for validating ideas before committing to final hardware architecture
- Design reviews are extremely valuable and can identify missing hardware before manufacturing
- Proper decoupling and power stability are critical in embedded system design
- Learning 4-layer PCB routing significantly improved understanding of professional PCB workflows
- Real engineering development is highly iterative and improves through debugging, redesign, and refinement
- This project provided practical experience with schematic capture, PCB routing, manufacturing preparation, debugging, and embedded system integration